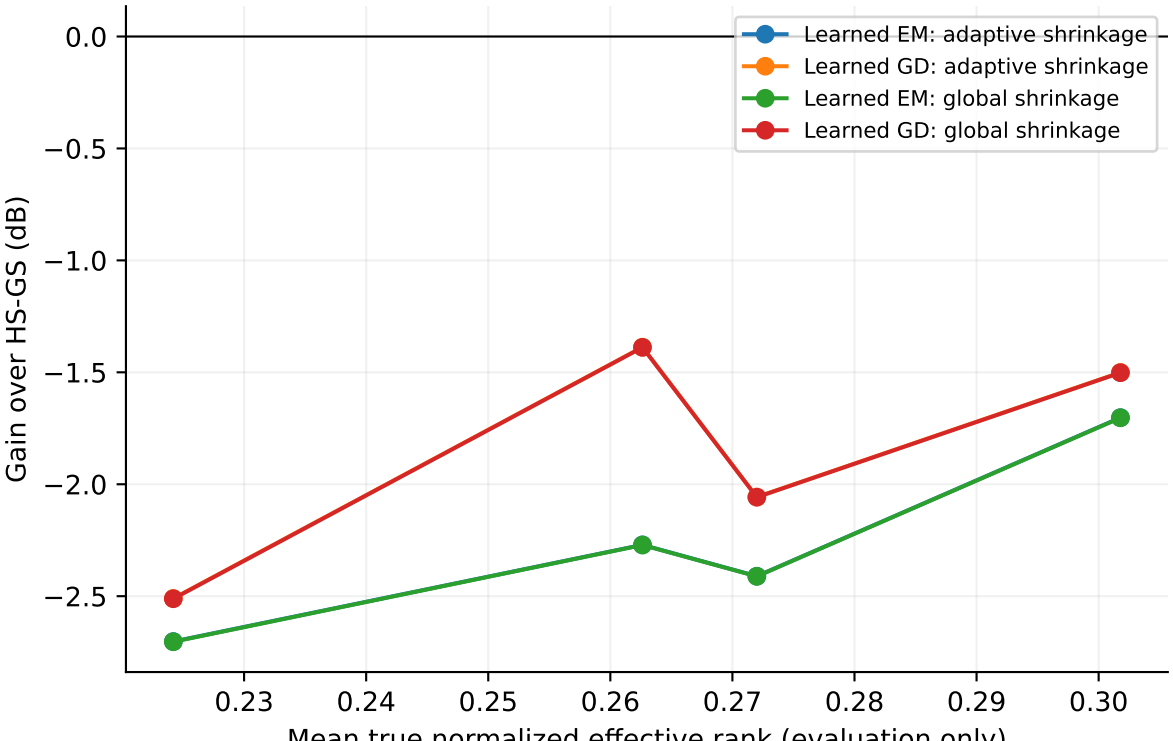
Pilot complexity diagnostic: four equal-count bins
Same 3–7-path distribution; not a complexity-transfer test
Learned EM: adaptive shrinkage
Learned GD: adaptive shrinkage
Learned EM: global shrinkage
Learned GD: global shrinkage
Gain over HS-GS (dB)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0.23
0.24
0.25
0.26
0.27
0.28
0.29
0.30
Mean true normalized effective rank (evaluation only)